

15.076 Project Proposal Guidelines

Due: March 30, 2026, 11:59 pm

Welcome to the project guidelines for Analytics for a Better World. In this project, you will practice applying the analytical methods and concepts you have learned in this course to an exciting and real-world problem related to the Sustainable Development Goals. Working in teams of two is encouraged, but if you prefer working alone, that's perfectly fine too. If you want to work in teams of more than two, please seek permission from the teaching team.

For the project, you can either try a new approach to a problem you studied in this course or choose an issue you are passionate about and explore how analytics can help solve it. To recap, we have discussed **various methods** (including linear, integer, robust optimization, heuristics, path-finding algorithms, and soon linear regression, logistic regression, dynamic model fitting, optimal classification trees, and deep learning). We have also seen **how analytics methods have been applied** to positively impact domains such as the World Food Program, dike height planning, and COVID-19. Please note that these lists are non-exhaustive – Refer to your notes and the lectures/recitations for more details! Now, it's your turn to make an impact!

Resources: You can reuse some of the ideas developed in recitations or homework assignments. Please also feel free to reach out to the teaching team for feedback or clarification at any time. You may also find the following websites helpful for getting some inspiration:

- <https://sdg-tracker.org/>,
- <https://sdgs.un.org/goals>, and
- <https://www.ai-for-sdgs.academy/casebase>.

Be creative and develop innovative solutions to the problems you are interested in. We are excited to see what you come up with!

Project deliverables: You will need to submit a **project proposal** when you return from Spring Break. The teaching team will then review your proposals and respond with feedback within 1-2 weeks. Near the end of the semester, you will present your project to the class during one of the final two lectures, in a 10-minute presentation (All members in each team are required to participate in the presentation). You will be required to turn in your finalized slide decks on Canvas before your presentations, and submit an 8-page report summarizing the methodology and main findings of your project.

Please refer to the subsequent pages for more details on each of the project deliverables (including required components for each deliverable, deadlines, and page/time limits).

1 Deadlines

- **Proposal** must be submitted on Canvas **by Monday, 3/30, 11:59 pm.**
- **Presentation slides** must be submitted on Canvas **by Friday, 5/1, 11:59 pm.**
- **8-page report** must be submitted on Canvas **by Friday, 5/8, 11:59 pm.**
- **In-class presentations** will be held **on 5/4, 5/6, and 5/11.** All members of the team must present, and the presentation should last about 10 minutes.

2 Proposal

Your proposal should be concise and **not exceed one page in length.** It should include:

- Name(s) and email(s) of team member(s).
- Problem summary, including the problem you want to solve and why it is important.
- Any data you plan on using and if you have already acquired it. Note: Using data is extremely important and will often be the project bottleneck. If you are struggling to find data, please let us know early on!
- How you plan to address the problem using tools from the class.

You are welcome to discuss the proposal with the teaching team. The project proposal will be graded Pass/Fail. Everybody should pass after feedback and potential adjustments.

3 Report

You will need to submit a report of up to 8 pages (excluding appendices). Your report should present and motivate the problem you are trying to solve clearly, describe your methodology, and summarize your main results. Your report should follow the guidelines of typical computer science conferences (e.g., ICML, AAAI, NeurIPS).

We will evaluate the success of your project based on the following criteria:

- **Strengths:** Typical criteria include soundness of the claims (theoretical grounding, empirical evaluation), significance and novelty of the contribution, and relevance to the community.
- **Weaknesses:** Your capacity to explain the limitations of your work, along the same axes as above.
- **Correctness:** Are the claims and method correct? Is the empirical methodology correct?
- **Clarity:** Is the paper well written? Is your method well-designed and explained?
- **Relation to prior work:** Did you explain how your contributions differ from previous work?
- **Reproducibility:** Are there enough details to reproduce the major results of this work?

Teams with > 1 member should also include a description of each team member's contributions.

4 In-Class Presentation

Details will be announced in April. You (or your group) will be assigned a slot to present on 5/4, 5/6 or 5/11, but will be required to attend and participate in all presentation sessions.